APPLIED TIME SERIES ANALYSIS · COURSE PROJECT

AQI Forecasting Lab

Forecasting Mumbai's daily Air Quality Index with classical time series models, from raw CPCB bulletins to a deployed forecasting dashboard.

AR · MA · ARMA · ARIMA · SARIMA

ADF + KPSS

MSTL decomposition

AIC / BIC

FastAPI + Plotly

PRESENTED BY

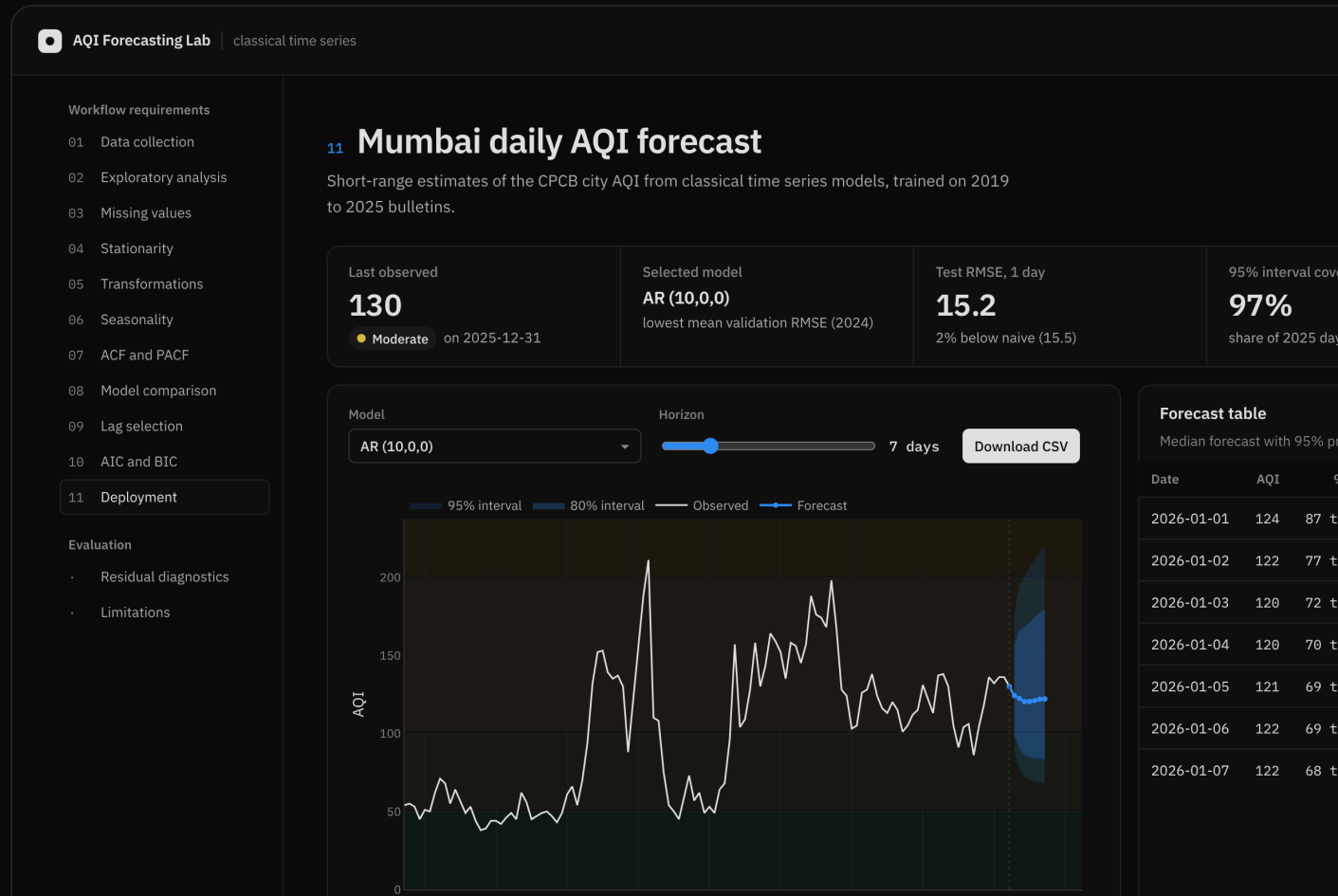
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Mumbai (primary) and Delhi · CPCB data 2015–2025

github.com/Anumay1231/aqi-forecasting-lab



THE BRIEF

Model deployment workflow: 11 required steps

Every step below is implemented in the pipeline and shown with real numbers in the dashboard. The rest of this deck walks through them in order.

01	Data collection	CPCB bulletins, 471k rows
02	Exploratory data analysis	line, scatter, histogram, QQ, stats
03	Dealing with missing values	3 methods tested
04	Check stationarity, ADF test	ADF + KPSS, 5 series
05	Transformations if not stationary	log, d = 1, D = 1
06	Seasonality, decomposition	MSTL: weekly + annual
07	ACF, PACF plots	log AQI and Δ log AQI
08	Identify AR, MA, ARMA, ARIMA, SARIMA	all five fitted
09	Identify lags	75 candidate orders
10	AIC and BIC criteria	per family + out-of-sample
11	Deploy the model	localhost dashboard + API
+	Evaluation (beyond the brief)	2025 hold-out, residual tests

CHOOSING THE DATA

From the suggested topics, we picked air pollution

Suggested data ideas

GDP	annual/quarterly, too few points
Population	smooth trend, little to model
Interest rate	policy steps, not stochastic
Currency	near random walk
Energy: crude oil, gas, electricity, solar	market-driven shocks
Bitcoin	random walk, regime changes
Air pollution, CO_2, temperature, rainfall	**chosen**
Sales forecasting	no open long series
Company financials, revenue, profit	quarterly, short

Why daily AQI suits classical time series

- **Long and frequent:** one value per day for 10+ years, about 2,500 usable days per city.
- **Real structure to find:** strong persistence (lag-1 correlation 0.91), a weekly cycle and a large annual monsoon cycle.
- **All five model families make sense:** autoregression, shocks that fade, and seasonality.
- **Official and open:** published every day by the Central Pollution Control Board (CPCB).
- **Useful output:** a 1–7 day AQI forecast with health category is something people act on.

Cities. Mumbai is the primary city. Delhi is run through the same pipeline as a contrast: much higher pollution and a stronger seasonal pattern.

STEP 01

01 Data collection

CPCB National AQI daily bulletins, parsed into one CSV by UrbanEmissions.info (GitHub: urbanemissionsinfo/AQI_bulletins, GPL-3.0).

471,015
rows, all cities

297
cities covered

2015–2025
1 May 2015 to 31 Dec 2025

3,614
Mumbai days with a bulletin (of 3,787)

What one row holds

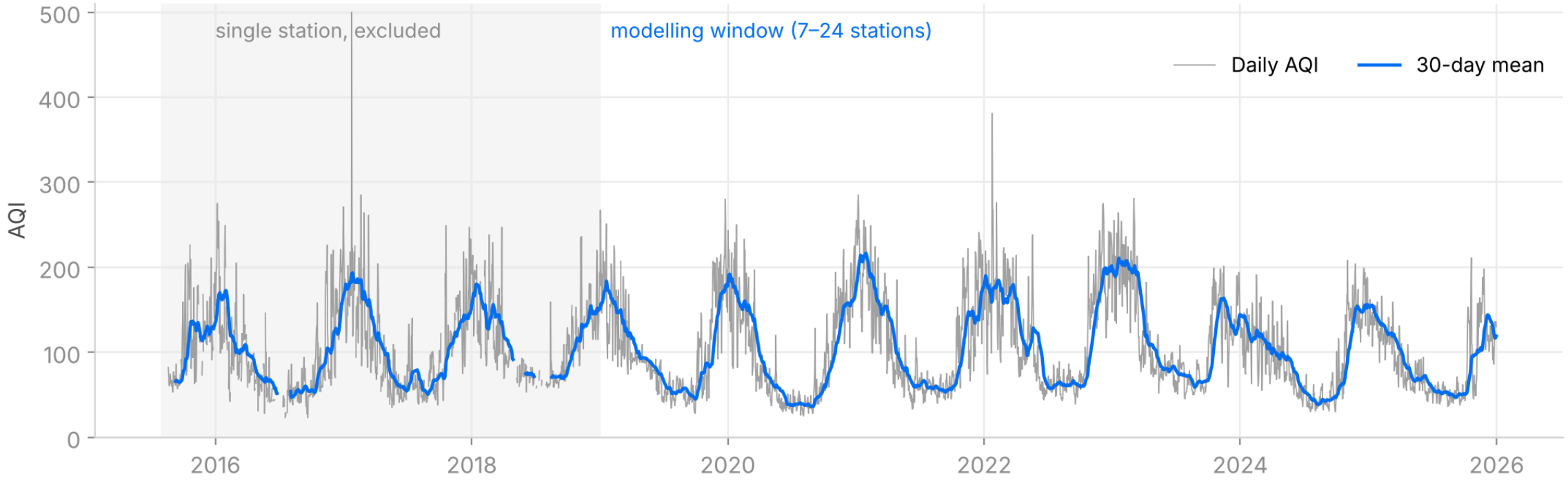
- Date, city, city AQI (0–500, average of all stations), number of stations, prominent pollutant, air quality category.
- No duplicate city-date rows and no zero AQI values in Mumbai's series.

Why modelling starts in 2019

- Until 2018 Mumbai had **one station**, so the "city" AQI was a single site. In 2019 it rose to 7, and to 24 by 2024.
- Using 2019–2025 keeps the series comparable and drops 168 of the 173 missing days.

Source: cpcb.nic.in/AQI_Bulletin.php

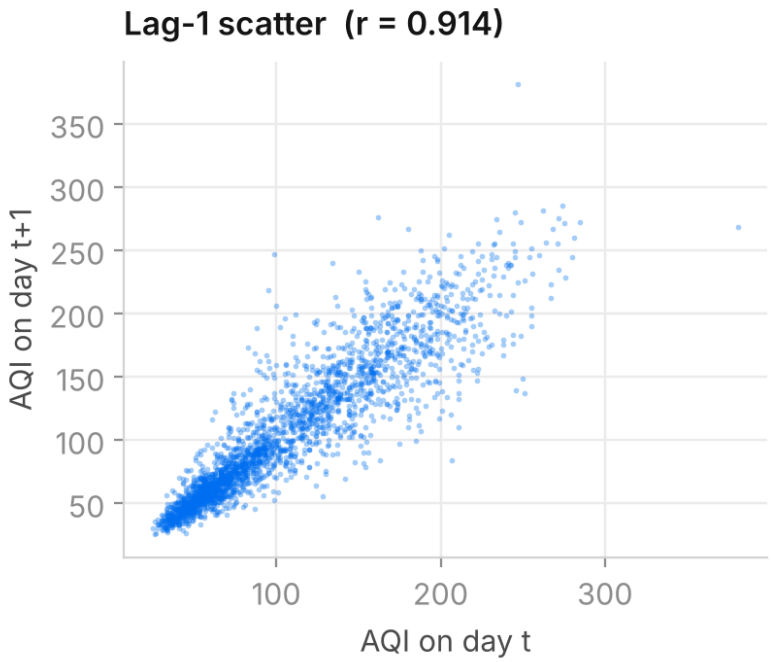
02 Line graph: ten years of Mumbai AQI

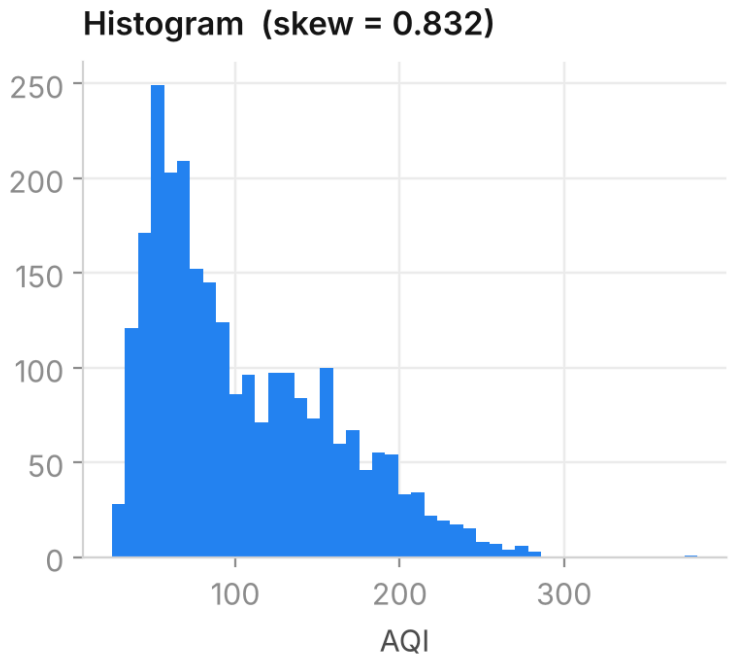


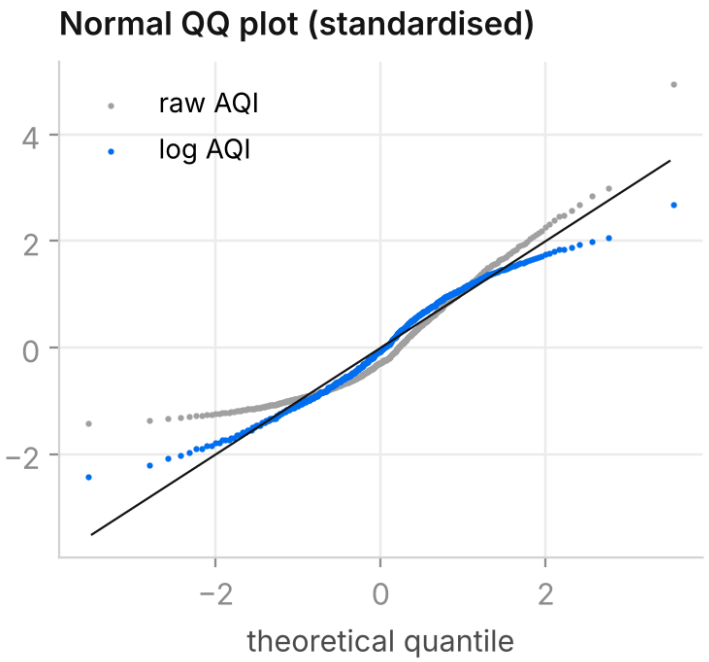
- Clear **annual cycle**: poor air every winter, clean air every monsoon.
- No long-run upward or downward trend after 2019.
- Day-to-day swings are large in winter and small in the monsoon: variance rises with the level.
- That points to a **log transform** (step 05).

Grey band: single-station years, excluded from modelling

Scatter plot, histogram and QQ plot







Scatter (lag 1)

Today's AQI predicts tomorrow's closely: r = **0.914**. At lag 7 r = 0.755 and at lag 365 r = 0.648. Autoregressive models should work.

Histogram

Right-skewed (skewness **0.832**): many clean days, a long tail of bad ones. After log, skewness falls to **0.012**.

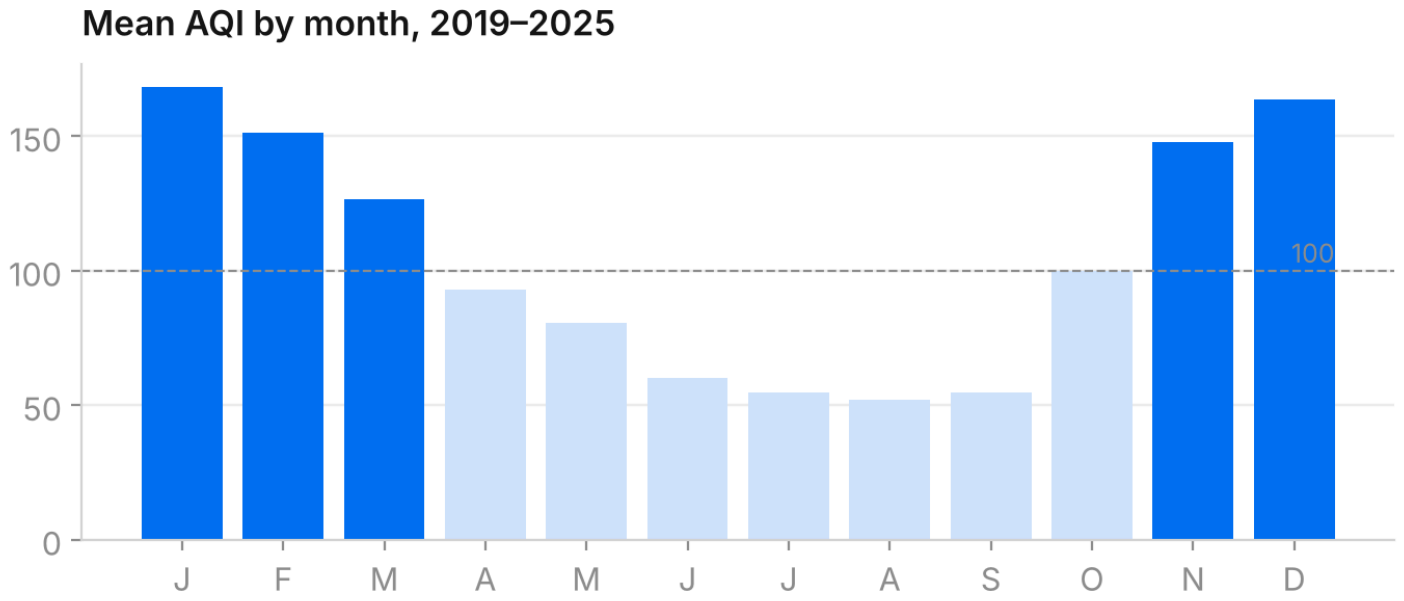
QQ plot

Raw AQI bends away from the normal line at both ends. Log AQI follows it much more closely.

STEP 02 · EXPLORATORY DATA ANALYSIS

Descriptive statistics and the seasonal profile

STATISTIC	MUMBAI AQI
Days	2,557
Mean	104.2
Std deviation	54.9
Minimum	25
25th percentile	59
Median	89
75th percentile	141
Maximum	381
Skewness (raw / log)	0.832 / 0.012
Excess kurtosis	0.007

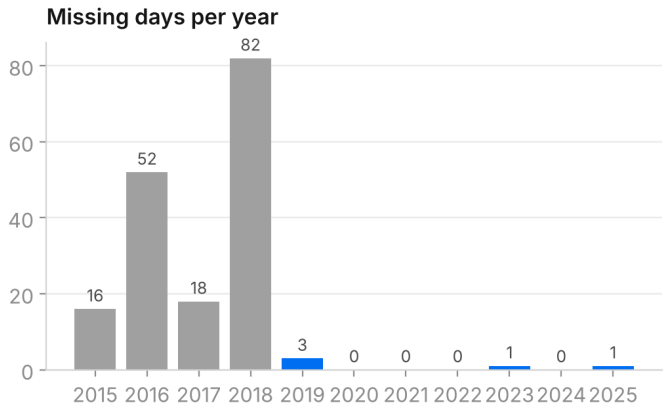


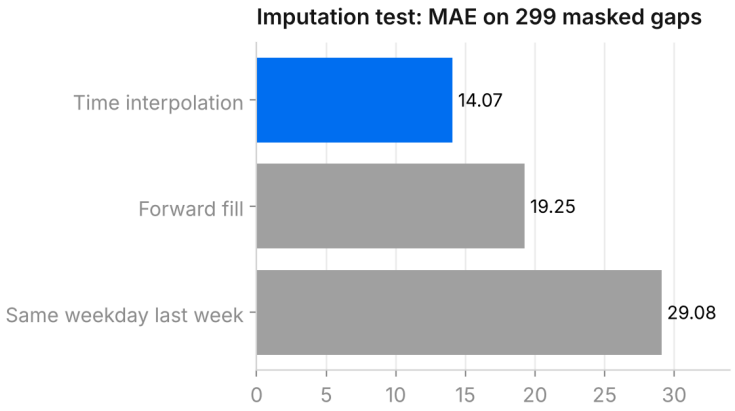
- January averages **168** (Moderate); August **52** (Satisfactory), three times cleaner.
- Kruskal-Wallis across months: p ≈ 0, a real seasonal effect. Across weekdays: p = 0.99, no meaningful weekly effect in the means.

AQI categories: 0–50 Good, 51–100 Satisfactory, 101–200 Moderate, 201–300 Poor

STEP 03

03 Dealing with missing values





173
missing days, full history (4.57%)

5
missing in 2019–2025 (0.20%)

2 days
longest gap in the window

How we filled them

- Rebuilt a full daily calendar so gaps are explicit.
- Tested three fill methods by **hiding 299 known values** and measuring the error.
- **Time interpolation won** (MAE 14.1 vs 19.3 forward fill, 29.1 same weekday last week).
- After filling: 0 missing values.

Missing dates in the window: 28 Feb, 1 Mar and 2 Apr 2019, 21 Sep 2023, 1 Jan 2025.

STEP 04

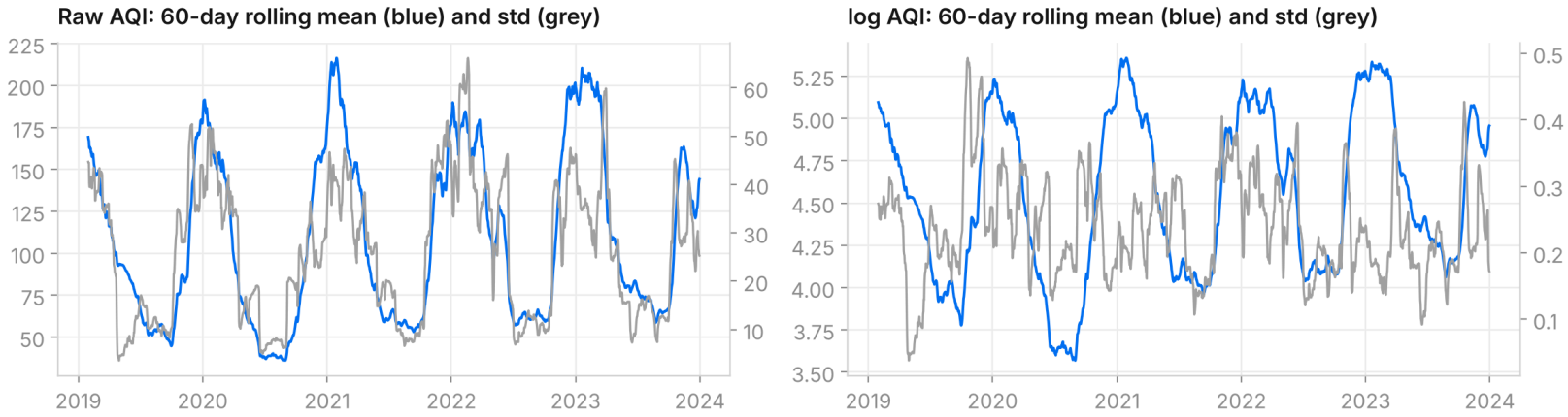
04 Check stationarity: ADF and KPSS tests

ADF: H_0 = unit root (non-stationary). KPSS: H_0 = stationary. Using both guards against either test's blind spots.

SERIES	ADF STAT	ADF P	KPSS STAT	KPSS P	VERDICT
AQI	-3.64	0.005	0.218	≥ 0.10	stationary
log AQI	-3.46	0.009	0.283	≥ 0.10	stationary
Δ log AQI (d = 1)	-20.27	<0.001	0.060	≥ 0.10	stationary
Δ_7 log AQI (D = 1, s = 7)	-10.16	<0.001	0.067	≥ 0.10	stationary
Δ Δ_7 log AQI	-15.62	<0.001	0.011	≥ 0.10	stationary

Result. The level series is already stationary at 5% on both tests: it returns to a seasonal mean and has no trend. So ARMA-type models (d = 0) are valid; d = 1 is still tried in ARIMA for comparison.

Rolling std of **raw** AQI tracks its mean (r = 0.81); for **log** AQI that link drops to 0.29.



STEP 05

05 Transformations

1 · Log transform (variance)

- Box-Cox estimated **λ = 0.03**, effectively 0, so we use the simpler `log(AQI)`.
- Mean–std correlation: **0.81 → 0.29**. Skewness: **0.83 → 0.01**.
- Forecasts come back with `exp()` (the median AQI); interval ends are transformed the same way and capped at 500.

2 · Differencing (mean)

- First difference `d = 1` and weekly seasonal difference `D = 1, s = 7` were tested in step 04.
- Not required for stationarity, but used inside the ARIMA and SARIMA candidates so the comparison is fair.

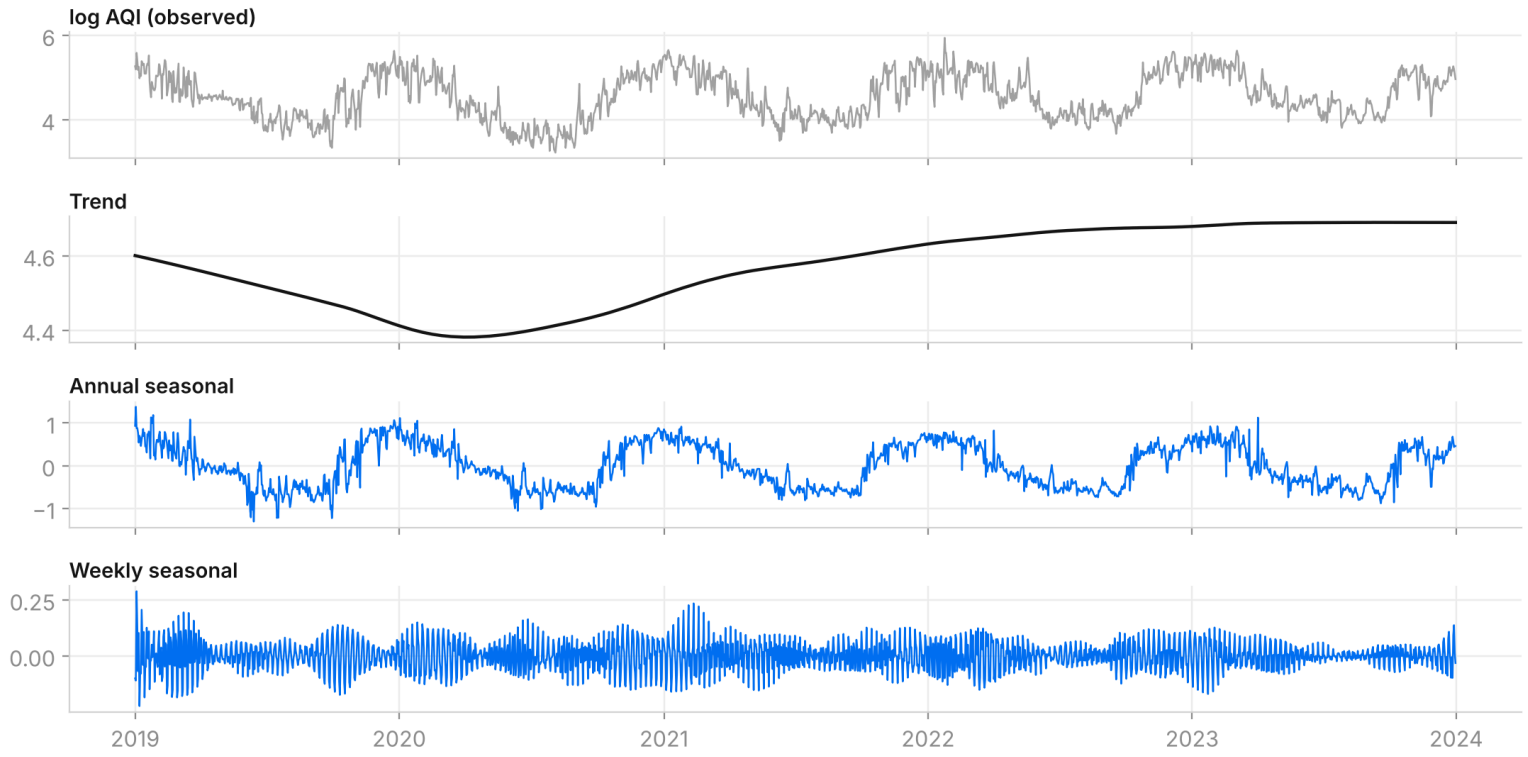
Effect of the log transform

MEASURE	RAW AQI	LOG AQI
Skewness	0.832	0.012
Mean–std correlation	0.813	0.290
QQ plot vs normal	bent tails	near line
Box-Cox λ		0.03 (≈ log)

All models are fitted to log AQI. This also keeps forecasts positive and makes the intervals wider on high-pollution days, which matches the data.

STEP 06

06 Seasonality and seasonal decomposition



MSTL splits log AQI into trend, a **weekly** (7-day) and an **annual** (365-day) seasonal part, and a residual.

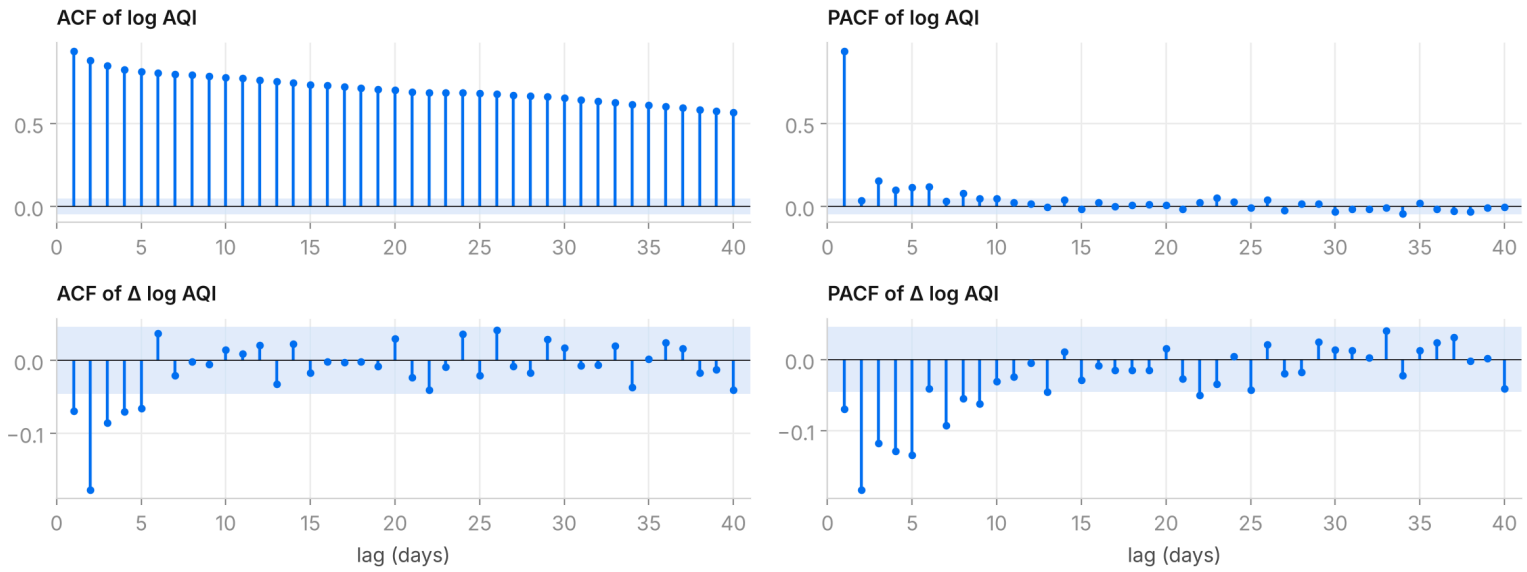
COMPONENT	STRENGTH
Annual seasonality	0.777
Trend	0.183
Weekly seasonality	0.045

- The annual cycle dominates (monsoon washout vs winter inversion).
- Weekly seasonality is weak, so SARIMA with s = 7 adds little.
- A 365-day SARIMA is impractical; the annual cycle is modelled with **Fourier terms** instead.

Strength = 1 − Var(resid) / Var(component + resid); 0 none, 1 strong

STEP 07

07 ACF and PACF plots



Reading the plots

- **ACF of log AQI** decays slowly: all 40 lags significant, a sign of strong persistence and the annual cycle.
- **PACF of log AQI** cuts off after a few lags: significant at 1, 3–6, 8–10 (and 23). This is an **AR signature**, pointing to p ≈ 10.
- After differencing, the ACF has a sharp spike at lag 1 and only 5 significant lags, which suggests an MA part with q ≈ 1–3.

Shaded band: 95% limits ±0.046 (n = 1,826).

STEP 08

08 Identify the model: AR, MA, ARMA, ARIMA, SARIMA

All five families from the brief were fitted, plus three reference models, so the choice is made on evidence.

FAMILY	WHAT IT ASSUMES	WHAT ACF/PACF SAID	BEST ORDER
AR(p)	today depends on the last p days	PACF cuts off, ACF decays: strong fit	`AR(10)`
MA(q)	today depends on the last q shocks	ACF does not cut off: poor fit	`MA(10)`
ARMA(p,q)	both	mixed decay in both plots	`ARMA(3,3)`
ARIMA(p,d,q)	ARMA on the differenced series	Δ ACF spikes at lag 1	`ARIMA(2,1,3)`
SARIMA	adds a weekly (s = 7) seasonal part	weak weekly peaks	`SARIMA(3,0,3)(0,0,1)[7]`
SARIMAX + Fourier	AR errors + sine/cosine annual cycle	strong annual seasonality (0.78)	`(3,0,0) + K = 2`
Baselines	Naive (tomorrow = today), seasonal naive (same weekday last week), Holt-Winters exponential smoothing		

Identification: the PACF cut-off plus slow ACF decay pointed to an AR model; the decomposition pointed to an annual seasonal term. Both were then confirmed by AIC/BIC and by forecast accuracy.

Fitted with statsmodels SARIMAX / ETSModel

STEP 09

09 Identify lags

A systematic search over 75 candidate orders

FAMILY	ORDERS TRIED	FITS
AR	`p = 1 … 14`	14
MA	`q = 1 … 10`	10
ARMA	`p, q ∈ {1,2,3}`	9
ARIMA	`d = 1, p ∈ 0…2, q ∈ 0…3`	15
SARIMA	`(P,D,Q)[7] variants`	7
SARIMAX + Fourier	`p ∈ 1…5, K ∈ 1…4`	20
Total		**75**

How the lags were chosen

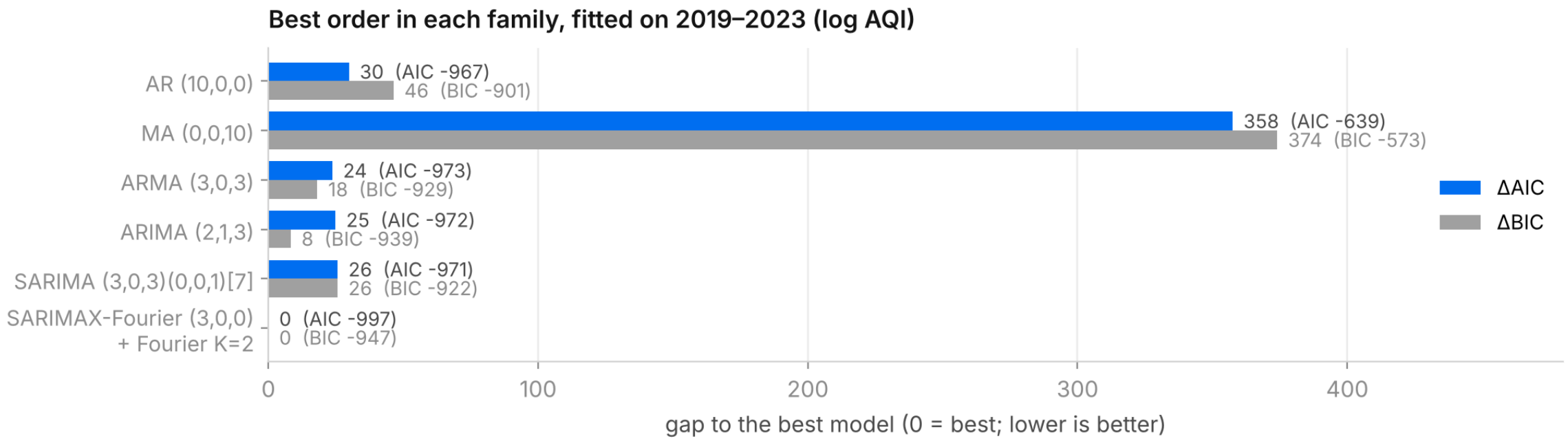
- **Start from the plots:** PACF lags 1–10 significant, so the AR search covers up to 14.
- **Rank by AIC** within each family (step 10).
- **Confirm on unseen data:** rolling 1, 3 and 7-day forecasts over 2024.
- **Check residuals:** Ljung-Box at lags 7, 14 and 28.

Selected lags: AR(10) uses the last 10 days. For Delhi the same search picked SARIMAX with K = 4 Fourier pairs.

Fitted on 2019–2023; nothing from 2024–2025 used in fitting

STEP 10

10 AIC and BIC criteria



SARIMAX + Fourier has the lowest AIC (−996.8) and BIC (−947.2): the annual cycle explains real variance.

MA(10) is far behind (AIC −639). AR, ARMA, ARIMA and SARIMA sit within 30 AIC points of each other.

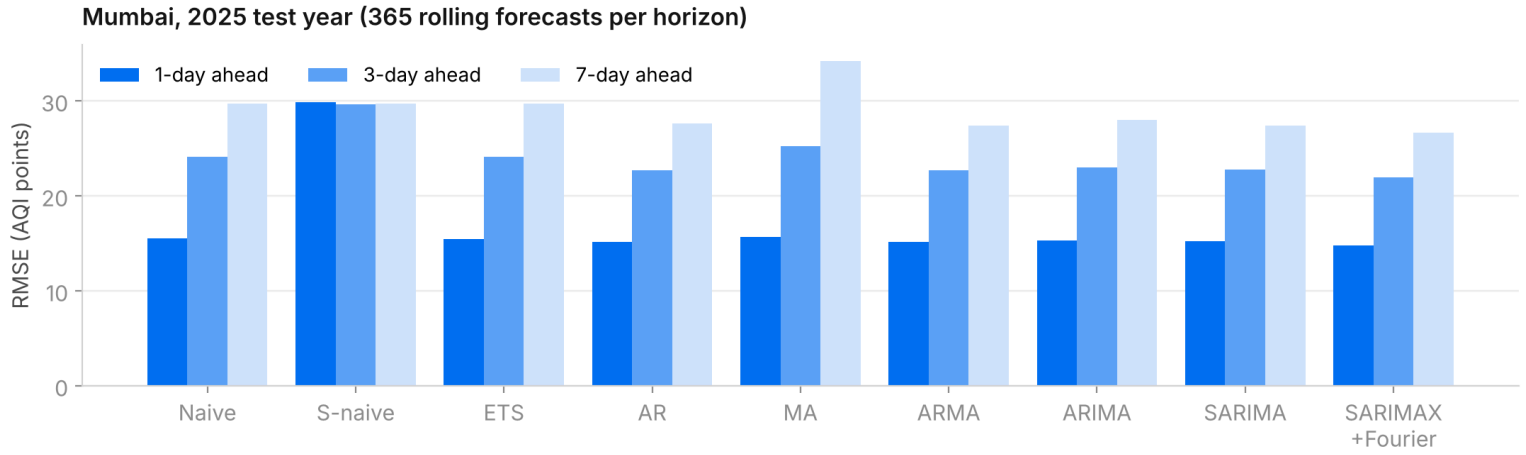
AIC is not comparable across different d, so the final pick uses **2024 forecast RMSE**: AR(10) was lowest (18.46 at 1 day).

AIC = 2k − 2 ln L · BIC = k ln n − 2 ln L

EVALUATION · 2025 HOLD-OUT

How accurate is it on a year no model has seen?

Split: train 2019–2023, validate 2024, test 2025 once. Every 2025 day gets a rolling forecast using only data up to the day before.

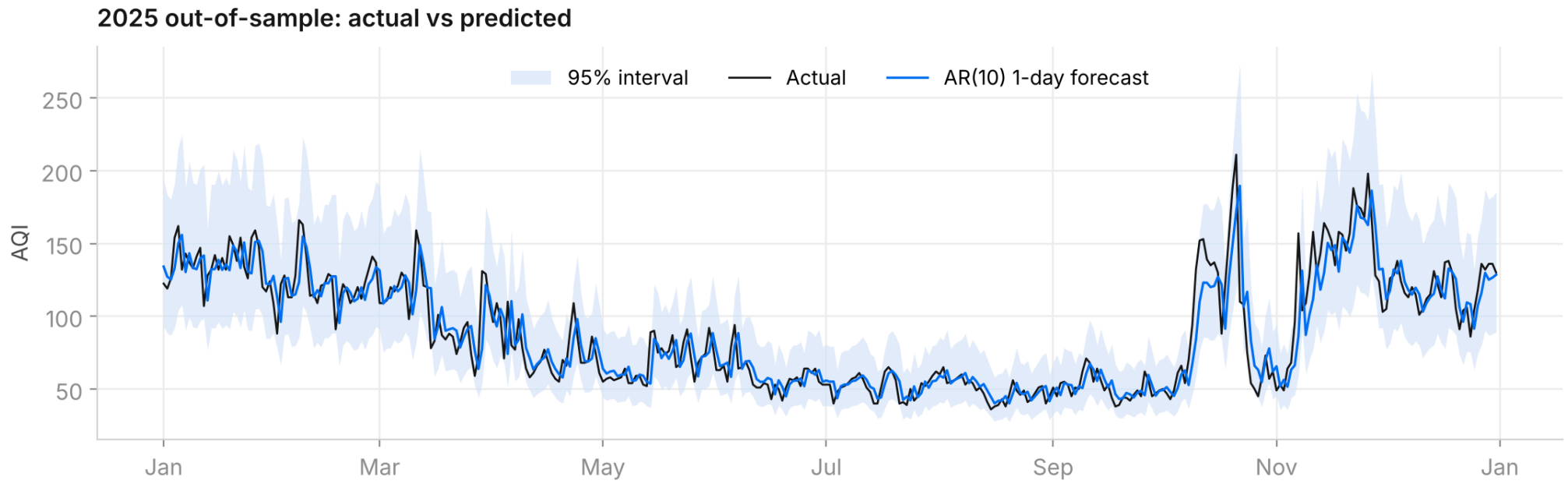


MODEL	RMSE 1D	RMSE 7D
Naive	15.53	29.71
AR(10)	15.15	27.58
ARMA(3,3)	15.17	27.38
ARIMA(2,1,3)	15.27	27.99
SARIMA	15.18	27.42
SARIMAX+F	14.80	26.64
MA(10)	15.64	34.19

AR(10), 1-day: MAE **10.3**, MASE 0.65, correct AQI category **83.6%** of days, 95% interval covers **97.3%**.

EVALUATION · 2025 HOLD-OUT

Actual vs predicted, and is the gain real?



Tracks the season

Forecasts follow the winter rise and monsoon fall, and the 95% band contains 97% of actual days.

Diebold-Mariano test vs naive

Mumbai: classical models are 2–10% better than naive, but **no p-value below 0.05** over one test year. Sudden weather-driven jumps dominate.

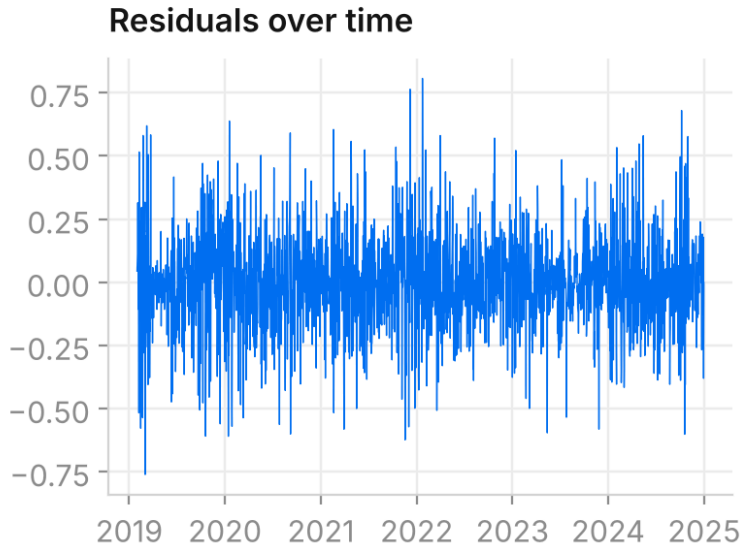
Delhi is different

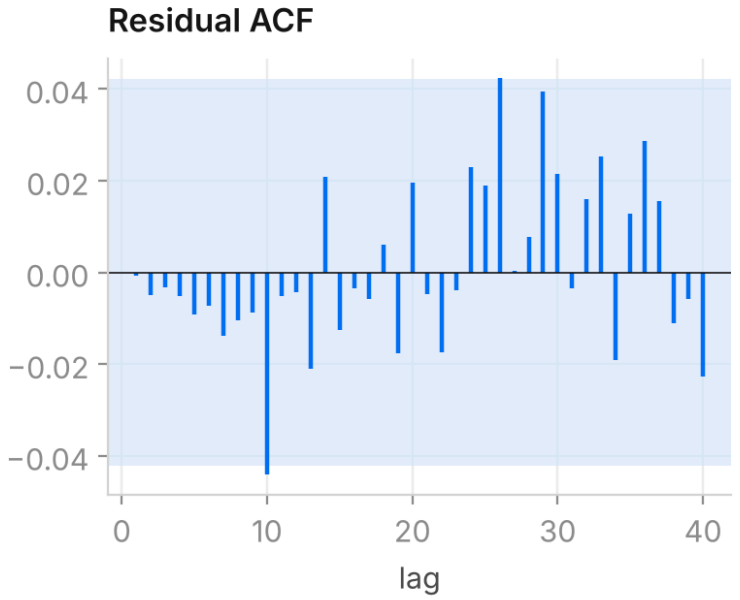
SARIMAX + Fourier beats naive at every horizon with **p < 0.0001** (slide 19).

AR(10), 1-day-ahead rolling forecasts, Jan–Dec 2025

EVALUATION · RESIDUAL DIAGNOSTICS

Residual diagnostics: is anything left to model?





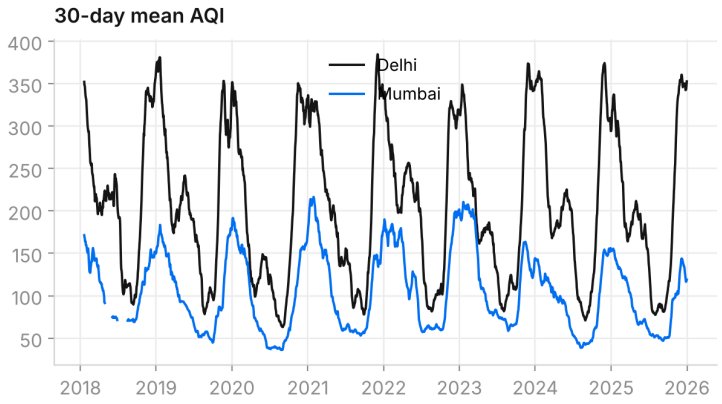


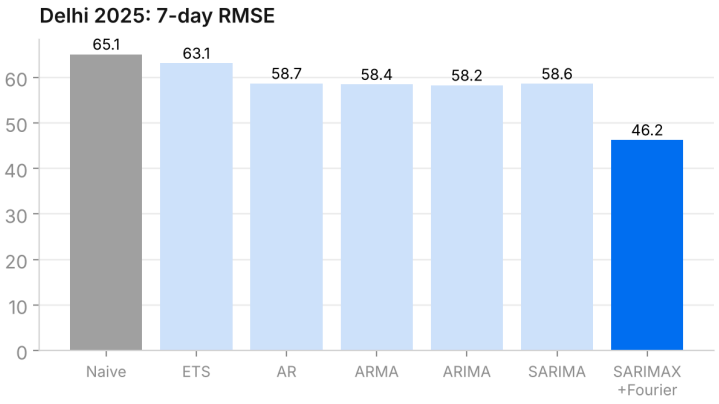
MODEL	LJUNG-BOX P, LAG 7	LAG 14	LAG 28	AUTOCORRELATION LEFT?	JARQUE-BERA
AR(10)	0.36	0.48	0.80	no: white noise	heavy tails (kurtosis 1.25)
ARIMA(2,1,3)	0.18	0.63	0.92	no	heavy tails
MA(10)	<0.001	<0.001	<0.001	yes: underfit	heavy tails

Residuals on the log scale, 2019–2024

SECOND CITY

Delhi: the same pipeline, a much stronger result



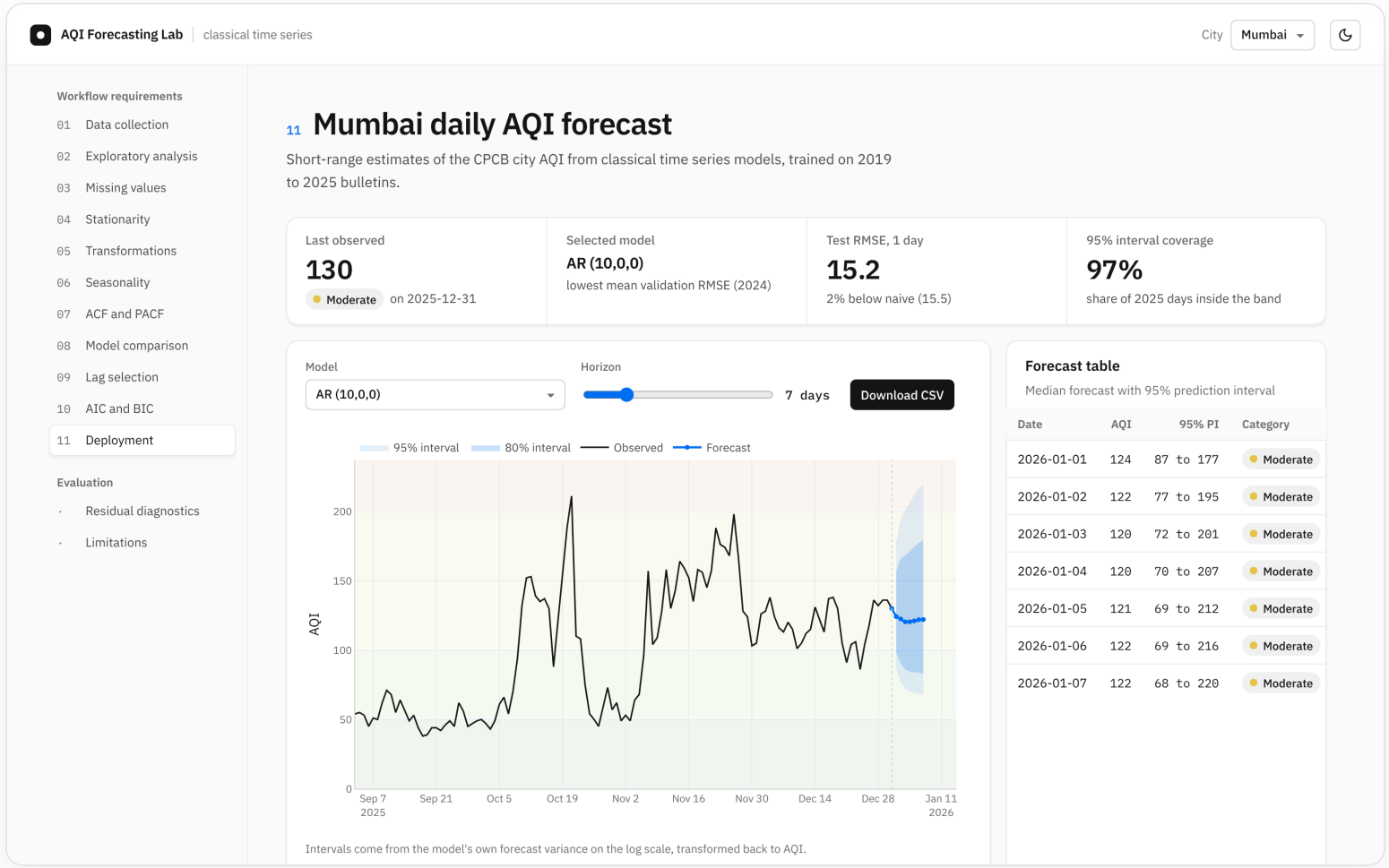


- Delhi averages **207** AQI (Mumbai 104) with a sharp, regular winter peak.
- Annual Fourier terms (K = 4) cut 7-day RMSE from **65.1 to 46.2**, a 29% gain.
- Diebold-Mariano vs naive: $p < 0.0001$ at 1, 3 and 7 days.
- Lesson: the stronger and more regular the seasonality, the more classical models add.

Delhi modelled from 2018; test year 2025

STEP 11

11 Deploy the model



- **Live forecasts:** pick city, any of 9 models, horizon 1–30 days.
- 80% and 95% prediction bands, AQI category colouring.
- Forecast table and **CSV download**.
- Every workflow step (01–10) has its own page section with the real charts and tests.
- Light and dark themes; works on a phone.

Next forecast: **1 Jan 2026 → AQI 124** (95% range 87–177), Moderate.

Screenshot of the running app at localhost:8000

STEP 11 · HOW IT IS BUILT

Architecture and how to run it

Pipeline → artifacts → API → dashboard

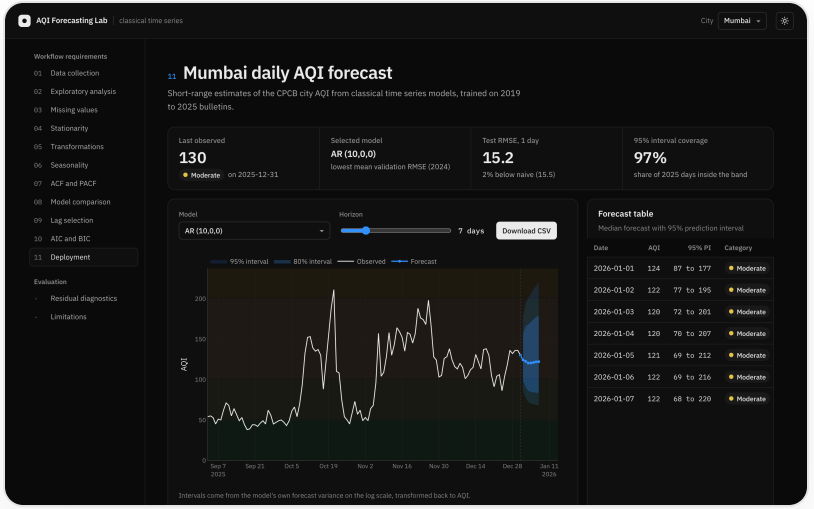
`run_pipeline.py`	runs steps 01–10 and fits all models for each city
`src/`	data loading, analysis, models, evaluation
`artifacts/*.json`	results and fitted parameters (no retraining on launch)
`app/server.py`	FastAPI: `/api/forecast`, `/api/forecast.csv`, `/api/city`
`app/static/`	HTML + CSS + JavaScript + Plotly charts
`tests/`	no data leakage, chronological splits, valid forecasts
`notebooks/`	the full analysis as a Jupyter notebook

Forecasts are computed on request from the saved model parameters; nothing is hard-coded. Each request takes well under a second.

Run it

```
Windows: double-click run.bat
macOS/Linux: ./run.sh
→ opens http://localhost:8000
```

First launch creates a private Python environment and installs 7 packages (1–2 min). Python 3.11+ is the only requirement.



CONCLUSION

Findings, limitations and next steps

Findings

- Mumbai AQI is stationary around a strong **annual cycle**, with heavy day-to-day persistence.
- **AR(10) on log AQI** is the selected model: best validation RMSE and white-noise residuals.
- Adding annual **Fourier terms** gives the best AIC and the best long-horizon accuracy.
- For Delhi the gain over naive is large and significant; for Mumbai it is small.

Limitations

- Only past AQI is used; weather (rain, wind) drives Mumbai's sudden swings.
- Residuals have heavy tails, so extreme days are under-predicted.
- One test year limits statistical power.

Next steps

- Add rainfall and wind as exogenous variables (SARIMAX).
- Extend to more cities from the same 297-city dataset.

SUMMARY

Requirement checklist

#	REQUIREMENT	WHERE IT IS DONE	
01	Data collection	CPCB bulletins, 471,015 rows; Mumbai 2019–2025	✓
02	EDA: line, scatter, histogram, QQ, descriptive stats	slides 5–7, dashboard section 02	✓
03	Missing values	5 days filled by time interpolation, chosen by test	✓
04	Stationarity, ADF test	ADF + KPSS on 5 series; level series stationary	✓
05	Transformations	log (Box-Cox λ = 0.03), d = 1, D = 1 tested	✓
06	Seasonality, decomposition	MSTL: annual strength 0.78, weekly 0.05	✓
07	ACF, PACF plots	log AQI and Δ log AQI, 40 lags	✓
08	AR, MA, ARMA, ARIMA, SARIMA	all five fitted, plus SARIMAX-Fourier and 3 baselines	✓
09	Identify lags	75 candidate orders; AR(10) selected	✓
10	AIC and BIC	per family, confirmed on 2024 and 2025 data	✓
11	Deploy the model	FastAPI + Plotly dashboard on localhost, one-click launch	✓